

Solution Requirements

Date	1 july 2026
Team ID	
Project Name	AI-Powered Student Mental Wellness Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely sign up and log in using email/password or university credentials. Passwords are encrypted before storage.	High
2	Authorization levels	Student users can access self-assessment, chatbot, mood tracking, and resources. Administrators can manage users, resources, reports, and analytics. Mental health professionals (optional) can	High

		view student requests with consent.	
3	External interfaces	Integrates with Gemini AI API for chatbot responses, email service for notifications, and cloud database (Firestore/MongoDB) for storing user data.	High
4	Transactions processing	Stores mood logs, chatbot conversations, assessment scores, and wellness activities in real time. Sends alerts or recommendations based on user responses.	High
5	Reporting	Generates dashboards showing mood trends, stress levels, chatbot usage, assessment history, and anonymous administrative statistics.	Medium
6	Business rules, etc.	Users complete a mental health assessment before receiving personalized recommendations. High-risk assessment scores trigger emergency support suggestions. Only authenticated users can access personal wellness data.	High

7	Compliance to laws or regulations	User data is stored securely following data privacy regulations. Personal information is encrypted, and user consent is required before sharing data.	High
8	<i>Other</i>	Provides personalized wellness tips, meditation exercises, daily mood reminders, emergency helpline information, and AI-generated coping strategies.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should respond quickly to user requests and chatbot queries.	Pages load in <2 seconds . Chatbot responses generated within 3 seconds .
2	Scalability	The platform should support increasing numbers of users without performance degradation.	Supports 10,000+ registered users and 1,000 concurrent users .
3	Security & Data Privacy	Protect user data using encryption, secure authentication, and restricted database access.	<i>Passwords hashed using bcrypt. Data encrypted using AES-256. HTTPS enabled.</i>
4	Reliability & Availability	The platform should remain available with minimal downtime and automatic	99.9% monthly uptime with automatic cloud backups.

5	Usability & Accessibility	The interface should be simple, responsive, mobile-friendly, and accessible to all	Mobile responsive with WCAG 2.1 AA accessibility compliance.
6	<i>Other</i>	Maintainability & Portability	Modular code architecture with API documentation. Compatible with Chrome, Firefox, Edge, Safari, and Android/iOS browsers.